

Jiwon Choi

ROBOTICS RESEARCHER · SLAM & SPATIAL PERCEPTION

Incheon, Republic of Korea

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Summary

I work on resilient state estimation for robots that must localize with degraded or minimal sensing. Building on KISS-IMU, our self-supervised inertial odometry framework, my current focus is proprioceptive state estimation for legged robots — explicitly modeling and correcting the environment-induced error in leg odometry using inertial, kinematic, and contact signals alone. I am an M.S.–Ph.D. integrated student at the Spatial AI and Robotics (SPARO) Lab at Inha University, Republic of Korea, advised by Prof. Younggun Cho.

Highlights

- First-author papers at IEEE ICRA: KISS-IMU (self-supervised inertial odometry, 2026) and the PoLaRIS maritime detection & tracking dataset (co-first author, 2025).
- KISS-IMU: ICRA 2026 Best Paper Award Finalist on Robot Perception, and Winner of the Outstanding WiRA Student Paper Award, ICRA 2026.
- 2nd Place, ICRA 2026 Hilti × Trimble SLAM Challenge (SLAM category).
- Samsung HumanTech Paper Award (Silver Prize, 2026); Graduate Research Award, Inha University (2026).
- Open-source author of KISS-IMU [📄 ★ 164] and PoLaRIS [📄 ★ 62].
- Reviewer for IEEE RA-L, IEEE ICRA, and IEEE/RSJ IROS (2025–2026).

Field of Interest

MY RESEARCH INTERESTS INCLUDE, BUT ARE NOT CONFINED TO:

- Self-supervised inertial learning and inertial odometry
- Robust and resilient state estimation under sensor degradation and limited supervision
- Proprioceptive state estimation for legged robots from inertial, kinematic, and contact sensing
- Explicit modeling and correction of environment-induced leg-odometry error
- Weak and cross-modal supervision for inertial and proprioceptive learning
- Multi-modal state estimation and SLAM using inertial, LiDAR, camera, and radar sensing

Education

Inha University

M.S.–PH.D. INTEGRATED IN ELECTRICAL AND COMPUTER ENGINEERING

- Advised by Prof. Younggun Cho, Spatial AI and Robotics (SPARO) Lab

Incheon, Republic of Korea

Mar. 2024 - Current

Inha University

B.S. IN ELECTRICAL ENGINEERING

Incheon, Republic of Korea

Mar. 2020 - Feb. 2024

Skills

Research Areas	SLAM, Self-supervised Inertial Learning, Proprioceptive State Estimation, Multi-modal Sensor Fusion
Programming	C++, Python, MATLAB
Frameworks & Tools	PyTorch, ROS/ROS2, Isaac Lab, Docker, Git, LaTeX
Languages	Korean (Native), English (Intermediate)

Research Project Experience

Dual Fisheye Camera-based Visual SLAM

SPARO Lab, Inha Univ.

2ND PLACE (OUT OF 62 TEAMS), ICRA 2026 HILTI X TRIMBLE SLAM CHALLENGE

2026

- Built dual fisheye camera-based Visual SLAM pipeline for the challenge benchmark.
- Placed 2nd among 62 teams with an average trajectory error below 0.1 m across 25 sequences.
- Keywords: Visual SLAM, Dual fisheye camera, Localization accuracy

Publications

Bold denotes the author; underline denotes co-first authors or co-corresponding authors.

INTERNATIONAL JOURNAL

1. Hogyun Kim, Geonmo Yang, **Jiwon Choi**, Juhui Lee, Seunghee Yun, Minho Lee, Gihyeon Lee, and Younggun Cho, "Obcalizer: Learning-Free and Stereo Vision-based Robust 3-DoF Object Localization and Tracking for Maritime Environments," *IEEE Trans. Intell. Transp. Syst. (T-ITS)*, 2026. (In preparation)
2. Hogyun Kim, **Jiwon Choi**, Juwon Kim, Geonmo Yang, Seokhwan Jeong, Hyungtae Lim, and Younggun Cho, "Commerge: Communication-Efficient, Robust and Fast LiDAR Map Merging Framework for Multi-robot Coordination in Resource-constrained Scenarios," *Int. J. Robot. Res. (IJRR)*, 2026.
3. Hogyun Kim, **Jiwon Choi**, Taehu Sim, Giseop Kim, and Younggun Cho, "Narrowing Your FOV with SOLiD: Spatially Organized and Lightweight Global Descriptor for FOV-constrained LiDAR Place Recognition," *IEEE Robot. Automat. Lett. (RA-L) with ICRA*, 2024.

INTERNATIONAL CONFERENCE

1. **Jiwon Choi**, Hogyun Kim, Geonmo Yang, Juhui Lee, and Younggun Cho, "KISS-IMU: Self-supervised Inertial Odometry with Motion-balanced Learning and Uncertainty-aware Inference," in *Proc. IEEE Int. Conf. Robot. Automat. (ICRA)*, 2026. (**Best Paper Award on Robot Perception, Finalist**)
2. **Jiwon Choi**^{*}, Dongjin Cho^{*}, Gihyeon Lee, Hogyun Kim, Geonmo Yang, Joowan Kim, and Younggun Cho, "Po-LaRIS Dataset: A Maritime Object Detection and Tracking Dataset in Pohang Canal," in *Proc. IEEE Int. Conf. Robot. Automat. (ICRA)*, 2025. (* Equally contribution)

PREPRINT

1. Hogyun Kim, **Jiwon Choi**, Juwon Kim, Geonmo Yang, Dongjin Cho, Hyungtae Lim, and Younggun Cho, "SKiD-SLAM: Robust, Lightweight, and Distributed Multi-Robot LiDAR SLAM in Resource-Constrained Field Environments," *arXiv preprint (ICRA Workshop on Field Robotics)*, 2025.

Honors & Awards

INTERNATIONAL

2026	Winner , Outstanding WiRA Student Paper Award, ICRA 2026	<i>Vienna, Austria</i>
2026	Finalist , Best Paper Award on Robot Perception, ICRA 2026	<i>Vienna, Austria</i>
2026	2nd Place (out of 62 teams) , Hilti x Trimble SLAM Challenge 2026, ICRA	<i>Vienna, Austria</i>
2025	Spotlight Talk , ICRA 2025 3rd Workshop on Field Robotics	<i>Atlanta, USA</i>

DOMESTIC

2026	Samsung HumanTech Paper Award (Silver Prize) , Samsung Electronics	<i>Seoul, S.Korea</i>
2026	Graduate Research Award , Inha University	<i>Incheon, S.Korea</i>
2025	Excellent Research Award , Korea Institute of ITS Conference	<i>Jeju, S.Korea</i>
2023	Excellent Research Award (Undergraduate Division) , Institute of Control, Robotics and System	<i>Samcheok, S.Korea</i>
2022	Alpha Project Competition (Grand Prize) , Inha University	<i>Incheon, S.Korea</i>
2022	Excellent Research Award , Korea Institute of ITS Conference	<i>Jeju, S.Korea</i>
2021	Address-Based Industry Creation Idea Contest (Grand Prize) , Ministry of the Interior and Safety	<i>Sejong, S.Korea</i>

Patents

GRANTED

2025	Domestic , Method and System for Self-Driving Car-based Road Management System	<i>Incheon, S.Korea</i>
2024	Domestic , Method and System for Fire Response Based on Smart Unmanned Self-Driving Platform	<i>Incheon, S.Korea</i>

APPLICATION

2025	International , Lightweight Robot Localization for LiDARs with Limited FOV	<i>Incheon, S.Korea</i>
2024	Domestic , Lightweight Robot Localization for LiDARs with Limited FOV	<i>Incheon, S.Korea</i>
2023	Domestic , A Method on the Efficient Multi-Floor Navigation Using Visual Marker and Floor Plan Map	<i>Incheon, S.Korea</i>

Academic Activities

Reviewer

- IEEE Robotics and Automation Letters (**RA-L 2025, 2026**)
- IEEE International Conference on Robotics and Automation (**ICRA 2026**)
- IEEE/RSJ International Conference on Intelligent Robots and Systems (**IROS 2026**)
- International Conference on Control, Automation and Systems (**ICCAS 2026**)